Every Minute Counts

A complete experiential learning study of Dijkstra's shortest path algorithm

Starting point: JAIN (Deemed-to-be University), School of Commerce, 44/4 District Fund Road, behind Smart Bazaar, Jayanagara 9th Block, Bengaluru 560041.

This report brings the complete study into one document: locations, road references, all graph costs, the algorithm, hand-worked traces, route alternatives, human journey observations, Google busy-period predictions, comparison charts, controlled scenarios, learning-session plan, interpretation, and source notes.

Study question

How does Dijkstra select the lowest-cost route in a small directed graph, how do distance and modeled driving time change the preferred path, and how do supplied human journey times compare descriptively with Google Maps busy-period predictions?

Main findings

Hospital	Shortest-distance route	Graph distance	Fastest-profile route	Graph profile time
C	A-B-C	1.497 km	A-B-C	2.18 min
D	A-B-C-D	2.695 km	A-B-C-D	3.61 min
L	A-B-C-L	4.608 km	A-B-C-L	6.14 min
M	A-M	0.779 km	A-M	1.08 min

The recorded trip was at or below Google's selected upper prediction for 7 of 11 comparable destinations. The median human-minus-Google difference was 0 min 37 sec, with the human trip shorter. Ragigudda bus stop (B) is excluded because its Google endpoint is provisional.

What the result means

Dijkstra is optimal for the graph and objective supplied to it. It does not make a larger road network, traffic forecast, or ambulance response system optimal. Here, edge costs are static car-profile estimates, while the human values are complete trips whose route, mode, and departure conditions were not supplied. The comparison is descriptive; it cannot measure Dijkstra accuracy or an effect on emergency response.

Scope and evidence

The graph contains 13 locations and 25 selected corridor pairs, represented by 50 independently weighted directed arcs. Human journey durations are supplied for all 12 destinations. There are no human route-choice, decision-time, or manual-calculation observations in the project data. No measured-route, GPS, hospital-entrance, or clinical-outcome claim is made.

Phase 1 - Locations and road references

A is the source for all trips. The hospital references are C, D, L and M. Stops, stations, a theatre, a mall and a temple provide additional graph structure. The graph points are mapped features snapped to a car-accessible road network; the listed offset is not added to route costs.

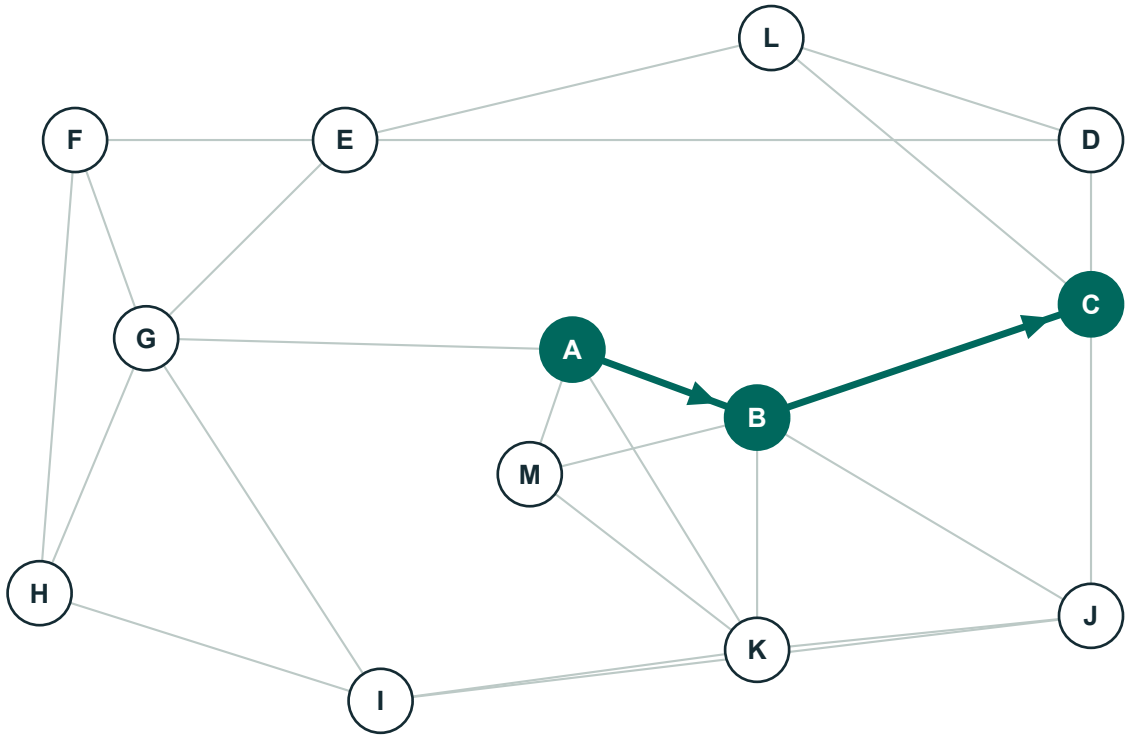
ID	Selected location	Mapped latitude, longitude	Road snap
A	JAIN University (source)	12.918126, 77.591874	29.8 m
B	Ragigudda bus stop	12.916799, 77.593385	4.1 m
C	Jayadeva hospital	12.918117, 77.599240	71.5 m
D	Sagar Hospitals	12.927948, 77.599850	41.6 m
E	Jayanagar TTMC	12.928171, 77.583868	2.4 m
F	Jayanagar metro bus stop	12.928771, 77.580345	3.8 m
G	RV Road station reference	12.921988, 77.580046	6.8 m
H	Banashankari bus stop	12.916631, 77.573069	4.1 m
I	Ranga Shankara	12.911608, 77.587080	23.7 m
J	Gopalan Innovation Mall	12.914339, 77.599454	39.8 m
K	Ragigudda Temple	12.914189, 77.593330	67.2 m
L	NIMHANS hospital reference	12.938632, 77.594410	200.5 m
M	Manipal Hospital, Jayanagar	12.917153, 77.591904	20.8 m

Manipal Hospitals, Jayanagar is the named branch. Its official address is 45/1, 45th Cross, Jayanagara 9th Block. The graph uses its mapped hospital reference snapped about 21 m to a road. The exact entrance is not verified. [S6]

Endpoint and feature notes

F is a bus stop named Jayanagara Metro Station, rather than a rail-platform coordinate. G is a station reference snapped to a road. H is the Banashankari bus stop. B refers to the Ragigudda bus stop, distinct from temple K. J is the mapped Gopalan Innovation Mall reference; the supplied “innovation center” trip is matched to it. L is a NIMHANS campus reference with the largest road snap, about 200 m. Hospital, temple, station and bus-stop entrances were not verified. [S1, S2]

Phase 1 - The 13-location teaching graph



Highlighted path: A-B-C. Lines are selected road-route corridors, not street shapes. Each pair has a separate cost in each direction. Crossing lines do not create graph junctions. [S2]

ID	Location	ID	Location
A	JAIN University (source)	H	Banashankari bus stop
B	Ragigudda bus stop	I	Ranga Shankara
C	Jayadeva hospital	J	Gopalan Innovation Mall
D	Sagar Hospitals	K	Ragigudda Temple
E	Jayanagar TTMC	L	NIMHANS hospital reference
F	Jayanagar metro bus stop	M	Manipal Hospital, Jayanagar
G	RV Road station reference		

Model boundaries

Each of the 25 displayed corridors uses a selected cell from the road-routing matrix. Intermediate roads and junctions remain hidden inside a corridor. Separately routed corridors may overlap or reverse direction physically; joining them can omit turn-history rules or create redundant travel. Therefore, “optimal” means optimal over this teaching graph, not over every legal Bengaluru street route. A production route model would use road-level segments, turn restrictions, vehicle access, live closures, and verified destination entrances.

Phase 1 - Complete directed corridor costs

Forward is first node to second node; reverse is second node to first. Distances are metres and durations are seconds from a static car-routing profile. The route API selects a profile route; these are not surveyed distances or observed segment times.

Corridor	Forward m	Reverse m	Forward s	Reverse s
AB	618.5	250.7	58.7	28.8
AK	829.9	821.7	83.3	83.9
AG	2053.7	2358.6	171.2	179.8
BC	878.5	1400.0	72.2	121.6
BK	624.2	571.0	59.1	55.1
BJ	1367.9	953.2	123.7	71.1
CD	1198.0	2107.0	85.8	202.2
CJ	1434.1	1067.1	118.0	95.0
CL	3110.9	3370.4	237.3	307.5
DE	2435.5	2368.2	234.8	221.1
DL	1996.7	2072.5	174.2	195.0
EF	771.2	454.3	68.3	51.6
EG	1571.5	1131.8	150.4	109.8
EL	2611.6	2306.0	229.4	236.0
FG	1228.6	817.5	119.4	71.9
FH	1974.3	1974.9	146.9	182.7
GH	1523.2	1511.9	120.5	172.6
GI	2196.0	1850.1	173.8	174.4
HI	2657.1	2137.0	245.0	160.3
IK	1227.2	1648.3	141.1	157.1
IJ	1959.0	2465.0	164.6	189.7
JK	1164.6	1017.4	103.8	108.6
AM	779.5	227.0	64.5	29.9
BM	702.9	541.3	45.9	36.6
KM	1273.9	752.8	101.0	61.2

The 25 corridor pairs form 50 directed arcs. Reverse costs are independently read from the road matrix, not copied from forward costs. The full 13-by-13 source-to-destination matrices follow in the appendices. [S2, S3]

Appendix A - Full road-routing distance matrix

All mapped nodes are included. Each cell is the road-route distance in metres from the row node to the column node. These matrix values are broader source data; only the 25 selected corridor pairs above are edges in the teaching graph.

source	A	B	C	D	E	F	G	H	I	J	K	L	M
A	0.0	618.5	1084.2	2282.3	2523.3	2824.1	2053.7	2340.6	1588.4	1573.6	829.9	2496.1	779.5
B	250.7	0.0	878.5	2076.6	2446.7	2747.5	1977.1	2264.0	1511.8	1367.9	624.2	2633.6	702.9
C	1387.8	1400.0	0.0	1198.0	3030.4	3324.8	3377.1	3664.0	2587.3	1434.1	1615.8	3110.9	2102.8
D	2024.2	2035.7	2107.0	0.0	2435.5	2724.0	3403.2	4304.2	3552.0	2547.3	2247.9	1996.7	2700.8
E	2376.3	2690.7	3156.4	2368.2	0.0	771.2	1571.5	2470.5	2198.4	3645.8	2902.1	2611.6	2149.3
F	2775.5	3089.9	3555.6	2783.9	454.3	0.0	1228.6	1974.3	2612.9	4045.0	3301.3	2595.5	2548.5
G	2358.6	2673.0	3138.8	3012.9	1131.8	817.5	0.0	1523.2	2196.0	3628.1	2884.4	3273.0	2131.7
H	2471.7	2786.1	3251.8	4449.8	2289.2	1974.9	1511.9	0.0	2657.1	3741.2	2997.5	4581.2	2244.7
I	2078.2	1925.9	2424.4	3572.6	2319.6	2616.5	1850.1	2137.0	0.0	1959.0	1227.2	3621.0	1851.2
J	1203.9	953.2	1067.1	2265.1	3399.9	3700.7	2930.3	3217.2	2465.0	0.0	1164.6	3489.3	1656.1
K	821.7	571.0	1069.4	2267.5	3017.6	3318.5	2548.1	2835.0	1648.3	1017.4	0.0	3107.0	1273.9
L	2503.2	2904.6	3370.4	2072.5	2306.0	2606.8	3737.3	4618.1	3865.9	4380.5	3116.0	0.0	3014.7
M	227.0	541.3	1007.1	2205.1	2446.1	2747.0	1976.6	2263.5	1511.3	1496.5	752.8	2609.8	0.0

Appendix B - Full road-routing duration matrix

All mapped nodes are included. Each cell is the static car-profile duration in seconds from the row node to the column node. Values are not human travel times or live traffic forecasts.

source	A	B	C	D	E	F	G	H	I	J	K	L	M
A	0.0	58.7	96.4	182.2	198.1	200.8	171.2	157.1	137.2	147.9	83.3	250.9	64.5
B	28.8	0.0	72.2	158.0	179.5	182.2	152.6	138.5	118.6	123.7	59.1	243.5	45.9
C	143.4	121.6	0.0	85.8	274.8	269.3	274.2	260.1	214.3	118.0	153.4	237.3	167.5
D	199.2	182.7	202.2	0.0	234.8	236.3	316.6	313.1	293.2	198.3	211.5	174.2	214.5
E	176.6	183.3	221.0	221.1	0.0	68.3	150.4	166.3	176.7	272.5	207.9	229.4	146.7
F	202.8	209.5	247.2	248.7	51.6	0.0	119.4	146.9	203.8	298.7	234.1	222.8	172.9
G	179.8	186.5	224.2	270.1	109.8	71.9	0.0	120.5	173.8	275.7	211.1	281.0	149.9
H	206.8	213.5	251.2	337.0	220.6	182.7	172.6	0.0	245.0	302.7	238.1	386.4	176.9
I	173.8	155.1	196.0	272.3	201.3	197.8	174.4	160.3	0.0	164.6	141.1	332.5	143.9
J	99.9	71.1	95.0	180.8	250.6	253.3	223.7	209.6	189.7	0.0	103.8	310.6	117.0
K	83.9	55.1	96.0	181.8	234.6	237.3	207.7	193.6	157.1	108.6	0.0	287.4	101.0
L	250.9	262.8	307.5	195.0	236.0	226.4	340.5	348.5	328.6	343.9	287.4	0.0	249.9
M	29.9	36.6	74.3	160.1	183.8	186.5	156.9	142.8	122.9	125.8	61.2	244.6	0.0

Phase 2 - Dijkstra method and correctness

For one selected destination and one non-negative edge weight, Dijkstra maintains a tentative cost for every node and a predecessor for each improved label.

Hand calculation: (1) Set A=0 and every other node to infinity. (2) Choose the unsettled node with the smallest tentative cost. (3) For each outgoing directed edge, add its weight to the settled cost. (4) Replace a neighbour's tentative cost and predecessor only when this new total is smaller. (5) Repeat until the destination is settled. Follow predecessor links backward to reconstruct the route.

Worked example - A to C by distance

Step	Settled	Cost (m)	Improved tentative labels
1	A	0.0	B=618.5 via A, G=2053.7 via A, K=829.9 via A, M=779.5 via A
2	B	618.5	C=1497.0 via B, J=1986.4 via B
3	M	779.5	No improvement
4	K	829.9	I=2478.2 via K, J=1847.3 via K
5	C	1497.0	No improvement

The predecessor chain is C ← B ← A, so the route is A-B-C. Its distance is 618.5 + 878.5 = 1,497.0 m. Its static duration on the same route is 58.7 + 72.2 = 130.9 s (2.18 min). The algorithm's distance display in the dashboard divides metres by 100 and truncates to two decimals; the report uses metres and kilometres.

Implementation and verification

The implementation uses a min-heap, tentative labels, predecessor pointers, stale-entry checks, and a settled set. Node-ID tie-breaking makes equal costs deterministic. Its usual binary-heap complexity is O((V+E) log V). Independent Floyd-Warshall calculations checked 169 source-destination pairs for distance and 169 for duration (338 cost comparisons); paths were reconstructed and checked. Boundary cases covered unreachable destinations, source equal to target, a zero-weight edge, equal-cost alternatives, and rejection of negative weights.

Code timing method

The graph is prebuilt. Each destination/objective uses 21 batches of 100 calls; the reported median and nearest-rank 95th percentile summarize the batch means in milliseconds. File access, network calls, manual tracing, and human work are excluded. These tiny local measurements illustrate computation, not city-scale performance.

Phase 2 - Route outcomes across all destinations

The teaching graph is compared with the direct OSRM road-profile reference for the same mapped points. The graph distance-optimal and time-optimal routes may differ. Direct OSRM is not a ground-truth check of Dijkstra; it uses a larger road graph and the same general profile-based assumptions.

ID	Distance-optimal route	Graph km	Fastest route	Graph min	Direct road km	Direct road min
B	A-B	0.619	A-B	0.98	0.619	0.98
C	A-B-C	1.497	A-B-C	2.18	1.084	1.61
D	A-B-C-D	2.695	A-B-C-D	3.61	2.282	3.04
E	A-G-E	3.186	A-G-E	4.68	2.523	3.30
F	A-G-F	2.871	A-G-F	4.05	2.824	3.35
G	A-G	2.054	A-G	2.85	2.054	2.85
H	A-G-H	3.577	A-G-H	4.86	2.341	2.62
I	A-K-I	2.478	A-K-I	4.01	1.588	2.29
J	A-K-J	1.847	A-B-J	3.04	1.574	2.47
K	A-K	0.830	A-K	1.39	0.830	1.39
L	A-B-C-L	4.608	A-B-C-L	6.14	2.496	4.18
M	A-M	0.779	A-M	1.08	0.779	1.07

A short route may leave the teaching graph because its corridor set is sparse. For example, L’s graph route is longer than its direct road reference because the graph exposes only selected connections. That is a modeling limitation, not a shortest-path algorithm error.

Shortest distance versus fastest modeled time

Objective	Route to J	Distance	Modeled duration
Distance	A-K-J	1847.3 m	191.9 s
Time	A-B-J	1986.4 m	182.4 s

A-K-J is shorter by distance. A-B-J is longer, but its profile duration is lower. “Shortest” and “fastest” are different optimization objectives and must not be treated as interchangeable.

Phase 2 - Local computation measurements

Median and p95 below are milliseconds per call, estimated from the mean of each batch. The machine and graph were local; these values are not a service-level performance measure.

ID	Distance route	Median ms	p95 ms	Time route	Median ms	p95 ms
B	A-B (0.62 km)	0.0033	0.0035	A-B (0.98 min)	0.0033	0.0058
C	A-B-C (1.50 km)	0.0078	0.0093	A-B-C (2.18 min)	0.0067	0.0076
D	A-B-C-D (2.69 km)	0.0123	0.0156	A-B-C-D (3.61 min)	0.0159	0.0177
E	A-G-E (3.19 km)	0.0175	0.0191	A-G-E (4.68 min)	0.0139	0.0171
F	A-G-F (2.87 km)	0.0124	0.0141	A-G-F (4.05 min)	0.0111	0.0133
G	A-G (2.05 km)	0.0106	0.0115	A-G (2.85 min)	0.0099	0.0122
H	A-G-H (3.58 km)	0.0179	0.0193	A-G-H (4.86 min)	0.0132	0.0156
I	A-K-I (2.48 km)	0.0114	0.0127	A-K-I (4.01 min)	0.0120	0.0141
J	A-K-J (1.85 km)	0.0104	0.0122	A-B-J (3.04 min)	0.0127	0.0138
K	A-K (0.83 km)	0.0070	0.0085	A-K (1.39 min)	0.0098	0.0145
L	A-B-C-L (4.61 km)	0.0186	0.0250	A-B-C-L (6.14 min)	0.0160	0.0199
M	A-M (0.78 km)	0.0061	0.0067	A-M (1.08 min)	0.0063	0.0082

The graph is small enough that calculation time is negligible beside route choice, data quality, travel, and endpoint validation. These timings do not compare Dijkstra with a live navigation service.

Phase 2 - Alternative routes to the hospitals

The dashboard lists up to five simple paths per selected destination, ranked by the chosen objective. The list below records the five lowest-cost routes by distance and by modeled time for each hospital. Distances shown are teaching-graph sums; human journey times are separate complete trips.

Hospital	Ranked by	Route	Path	Distance	Profile time
C	Distance	1	A-B-C	1.497 km	2.18 min
C	Distance	2	A-M-B-C	2.199 km	2.89 min
C	Distance	3	A-K-B-C	2.279 km	3.51 min
C	Distance	4	A-K-J-C	2.914 km	4.78 min
C	Distance	5	A-M-K-B-C	2.982 km	4.22 min
C	Modeled time	1	A-B-C	1.497 km	2.18 min
C	Modeled time	2	A-M-B-C	2.199 km	2.89 min
C	Modeled time	3	A-K-B-C	2.279 km	3.51 min
C	Modeled time	4	A-M-K-B-C	2.982 km	4.22 min
C	Modeled time	5	A-B-J-C	3.054 km	4.62 min
D	Distance	1	A-B-C-D	2.695 km	3.61 min
D	Distance	2	A-M-B-C-D	3.397 km	4.32 min
D	Distance	3	A-K-B-C-D	3.477 km	4.94 min
D	Distance	4	A-K-J-C-D	4.112 km	6.21 min
D	Distance	5	A-M-K-B-C-D	4.180 km	5.65 min
D	Modeled time	1	A-B-C-D	2.695 km	3.61 min
D	Modeled time	2	A-M-B-C-D	3.397 km	4.32 min
D	Modeled time	3	A-K-B-C-D	3.477 km	4.94 min
D	Modeled time	4	A-M-K-B-C-D	4.180 km	5.65 min
D	Modeled time	5	A-B-J-C-D	4.252 km	6.05 min
L	Distance	1	A-B-C-L	4.608 km	6.14 min
L	Distance	2	A-B-C-D-L	4.692 km	6.51 min
L	Distance	3	A-M-B-C-L	5.310 km	6.84 min
L	Distance	4	A-K-B-C-L	5.390 km	7.47 min
L	Distance	5	A-M-B-C-D-L	5.394 km	7.22 min
L	Modeled time	1	A-B-C-L	4.608 km	6.14 min
L	Modeled time	2	A-B-C-D-L	4.692 km	6.51 min
L	Modeled time	3	A-M-B-C-L	5.310 km	6.84 min
L	Modeled time	4	A-M-B-C-D-L	5.394 km	7.22 min
L	Modeled time	5	A-K-B-C-L	5.390 km	7.47 min
M	Distance	1	A-M	0.779 km	1.07 min
M	Distance	2	A-B-M	1.321 km	1.74 min
M	Distance	3	A-K-B-M	2.104 km	3.07 min

Hospital	Ranked by	Route	Path	Distance	Profile time
M	Distance	4	A-K-M	2.104 km	3.07 min
M	Distance	5	A-B-K-M	2.517 km	3.65 min
M	Modeled time	1	A-M	0.779 km	1.07 min
M	Modeled time	2	A-B-M	1.321 km	1.74 min
M	Modeled time	3	A-K-B-M	2.104 km	3.07 min
M	Modeled time	4	A-K-M	2.104 km	3.07 min
M	Modeled time	5	A-B-K-M	2.517 km	3.65 min

Phase 2 - Controlled corridor scenarios

The teaching scenario either removes corridor BC in both directions or triples only its static duration. Distances are unaffected by the duration change. These are controlled changes to the teaching graph; they do not simulate a measured congestion event or recompute all overlapping streets.

Target	Optimize	Scenario	Selected route	Distance	Profile time
C	distance	baseline	A-B-C	1.497 km	2.18 min
C	distance	BC corridor unavailable	A-K-J-C	2.914 km	4.78 min
C	distance	BC duration tripled	A-B-C	1.497 km	4.59 min
C	duration	baseline	A-B-C	1.497 km	2.18 min
C	duration	BC corridor unavailable	A-B-J-C	3.054 km	4.62 min
C	duration	BC duration tripled	A-B-C	1.497 km	4.59 min
D	distance	baseline	A-B-C-D	2.695 km	3.61 min
D	distance	BC corridor unavailable	A-K-J-C-D	4.112 km	6.21 min
D	distance	BC duration tripled	A-B-C-D	2.695 km	6.02 min
D	duration	baseline	A-B-C-D	2.695 km	3.61 min
D	duration	BC corridor unavailable	A-B-J-C-D	4.252 km	6.05 min
D	duration	BC duration tripled	A-B-C-D	2.695 km	6.02 min
J	distance	baseline	A-K-J	1.847 km	3.20 min
J	distance	BC corridor unavailable	A-K-J	1.847 km	3.20 min
J	distance	BC duration tripled	A-K-J	1.847 km	3.20 min
J	duration	baseline	A-B-J	1.986 km	3.04 min
J	duration	BC corridor unavailable	A-B-J	1.986 km	3.04 min
J	duration	BC duration tripled	A-B-J	1.986 km	3.04 min
L	distance	baseline	A-B-C-L	4.608 km	6.14 min
L	distance	BC corridor unavailable	A-G-E-L	5.797 km	8.51 min
L	distance	BC duration tripled	A-B-C-L	4.608 km	8.54 min
L	duration	baseline	A-B-C-L	4.608 km	6.14 min
L	duration	BC corridor unavailable	A-G-E-L	5.797 km	8.51 min
L	duration	BC duration tripled	A-G-E-L	5.797 km	8.51 min
M	distance	baseline	A-M	0.779 km	1.07 min
M	distance	BC corridor unavailable	A-M	0.779 km	1.07 min
M	distance	BC duration tripled	A-M	0.779 km	1.07 min
M	duration	baseline	A-M	0.779 km	1.07 min
M	duration	BC corridor unavailable	A-M	0.779 km	1.07 min
M	duration	BC duration tripled	A-M	0.779 km	1.07 min

The scenario lets students predict and test how Dijkstra responds to altered edge weights or connectivity. If a real street closure affects overlapping corridors, every affected road segment and junction must be rerouted before drawing a real-world conclusion.

Phase 2 - Full hand-worked hospital traces

Each trace shows every settled node until the target is settled, for both distance and modeled-time objectives. An “improvement” records the new tentative label and its predecessor. Distance is in metres; profile time is in seconds, with minutes shown in parentheses.

C - Jayadeva hospital

Step	Settled	Current cost	Improved tentative labels
1	A	0.0	B=618.5 via A, G=2053.7 via A, K=829.9 via A, M=779.5 via A
2	B	618.5	C=1497.0 via B, J=1986.4 via B
3	M	779.5	No label improved
4	K	829.9	I=2478.2 via K, J=1847.3 via K
5	C	1497.0	No label improved

Step	Settled	Current cost	Improved tentative labels
1	A	0.0 s (0.00 min)	B=58.7 s (0.98 min) via A, G=171.2 s (2.85 min) via A, K=83.3 s (1.39 min) via A, M=64.5 s (1.07 min) via A
2	B	58.7 s (0.98 min)	C=130.9 s (2.18 min) via B, J=182.4 s (3.04 min) via B
3	M	64.5 s (1.07 min)	No label improved
4	K	83.3 s (1.39 min)	I=240.4 s (4.01 min) via K
5	C	130.9 s (2.18 min)	No label improved

Reconstruction: A-B-C for both objectives. Distance 1,497.0 m; modeled duration 130.9 s.

Phase 2 - Full hand-worked hospital traces: D

D - Sagar Hospitals

Step	Settled	Current cost	Improved tentative labels
1	A	0.0	B=618.5 via A, G=2053.7 via A, K=829.9 via A, M=779.5 via A
2	B	618.5	C=1497.0 via B, J=1986.4 via B
3	M	779.5	No label improved
4	K	829.9	I=2478.2 via K, J=1847.3 via K
5	C	1497.0	D=2695.0 via C, L=4607.9 via C
6	J	1847.3	No label improved
7	G	2053.7	E=3185.5 via G, F=2871.2 via G, H=3576.9 via G
8	I	2478.2	No label improved
9	D	2695.0	No label improved

Step	Settled	Current cost	Improved tentative labels
1	A	0.0 s (0.00 min)	B=58.7 s (0.98 min) via A, G=171.2 s (2.85 min) via A, K=83.3 s (1.39 min) via A, M=64.5 s (1.07 min) via A
2	B	58.7 s (0.98 min)	C=130.9 s (2.18 min) via B, J=182.4 s (3.04 min) via B
3	M	64.5 s (1.07 min)	No label improved
4	K	83.3 s (1.39 min)	I=240.4 s (4.01 min) via K
5	C	130.9 s (2.18 min)	D=216.7 s (3.61 min) via C, L=368.2 s (6.14 min) via C
6	G	171.2 s (2.85 min)	E=281.0 s (4.68 min) via G, F=243.1 s (4.05 min) via G, H=291.7 s (4.86 min) via G
7	J	182.4 s (3.04 min)	No label improved
8	D	216.7 s (3.61 min)	No label improved

Reconstruction: A-B-C-D for both objectives. Distance 2,695.0 m; modeled duration 216.7 s.

Phase 2 - Full hand-worked hospital traces: L

L - NIMHANS hospital reference

Step	Settled	Current cost	Improved tentative labels
1	A	0.0	B=618.5 via A, G=2053.7 via A, K=829.9 via A, M=779.5 via A
2	B	618.5	C=1497.0 via B, J=1986.4 via B
3	M	779.5	No label improved
4	K	829.9	I=2478.2 via K, J=1847.3 via K
5	C	1497.0	D=2695.0 via C, L=4607.9 via C
6	J	1847.3	No label improved
7	G	2053.7	E=3185.5 via G, F=2871.2 via G, H=3576.9 via G
8	I	2478.2	No label improved
9	D	2695.0	No label improved
10	F	2871.2	No label improved
11	E	3185.5	No label improved
12	H	3576.9	No label improved
13	L	4607.9	No label improved

Step	Settled	Current cost	Improved tentative labels
1	A	0.0 s (0.00 min)	B=58.7 s (0.98 min) via A, G=171.2 s (2.85 min) via A, K=83.3 s (1.39 min) via A, M=64.5 s (1.07 min) via A
2	B	58.7 s (0.98 min)	C=130.9 s (2.18 min) via B, J=182.4 s (3.04 min) via B
3	M	64.5 s (1.07 min)	No label improved
4	K	83.3 s (1.39 min)	I=240.4 s (4.01 min) via K
5	C	130.9 s (2.18 min)	D=216.7 s (3.61 min) via C, L=368.2 s (6.14 min) via C
6	G	171.2 s (2.85 min)	E=281.0 s (4.68 min) via G, F=243.1 s (4.05 min) via G, H=291.7 s (4.86 min) via G
7	J	182.4 s (3.04 min)	No label improved
8	D	216.7 s (3.61 min)	No label improved
9	I	240.4 s (4.01 min)	No label improved
10	F	243.1 s (4.05 min)	No label improved
11	E	281.0 s (4.68 min)	No label improved
12	H	291.7 s (4.86 min)	No label improved
13	L	368.2 s (6.14 min)	No label improved

Reconstruction: A-B-C-L for both objectives. Distance 4,607.9 m; modeled duration 368.2 s. The mapped hospital reference snaps about 200 m to a road; entrance access is unverified.

Phase 2 - Full hand-worked hospital traces: M

M - Manipal Hospital, Jayanagar

Step	Settled	Current cost	Improved tentative labels
1	A	0.0	B=618.5 via A, G=2053.7 via A, K=829.9 via A, M=779.5 via A
2	B	618.5	C=1497.0 via B, J=1986.4 via B
3	M	779.5	No label improved

Step	Settled	Current cost	Improved tentative labels
1	A	0.0 s (0.00 min)	B=58.7 s (0.98 min) via A, G=171.2 s (2.85 min) via A, K=83.3 s (1.39 min) via A, M=64.5 s (1.07 min) via A
2	B	58.7 s (0.98 min)	C=130.9 s (2.18 min) via B, J=182.4 s (3.04 min) via B
3	M	64.5 s (1.07 min)	No label improved

Reconstruction: A-M for both objectives. Distance 779.5 m; modeled duration 64.5 s. The graph ends at a nearby road reference, not a verified entrance.

Phase 3 - Human journey times and Google predictions

The human values are actual journey durations supplied for trips from JAIN University. Google's prediction is the upper end of the saved busy-period driving estimate. The Google value is a forecast snapshot, not live traffic and not a guaranteed maximum. The human trip's route, transport mode, and departure conditions were not supplied.

ID	Destination	Human recorded	Google prediction	Difference	Reading
B	Ragigudda bus stop	6 min 23 sec	3 min	+3 min 23 sec	Provisional endpoint
C	Jayadeva hospital	9 min 48 sec	10 min	−0 min 12 sec	Human shorter
D	Sagar Hospitals	12 min 35 sec	14 min	−1 min 25 sec	Human shorter
E	Jayanagar TTMC	14 min 56 sec	14 min	+0 min 56 sec	Human longer
F	Jayanagar metro bus stop	18 min 34 sec	20 min	−1 min 26 sec	Human shorter
G	RV Road station reference	13 min 23 sec	14 min	−0 min 37 sec	Human shorter
H	Banashankari bus stop	25 min 59 sec	30 min	−4 min 01 sec	Human shorter
I	Ranga Shankara	13 min 21 sec	12 min	+1 min 21 sec	Human longer
J	Gopalan Innovation Mall	14 min 11 sec	14 min	+0 min 11 sec	Human longer
K	Ragigudda Temple	7 min 46 sec	4 min	+3 min 46 sec	Human longer
L	NIMHANS hospital reference	17 min 32 sec	20 min	−2 min 28 sec	Human shorter
M	Manipal Hospital, Jayanagar	7 min 19 sec	8 min	−0 min 41 sec	Human shorter

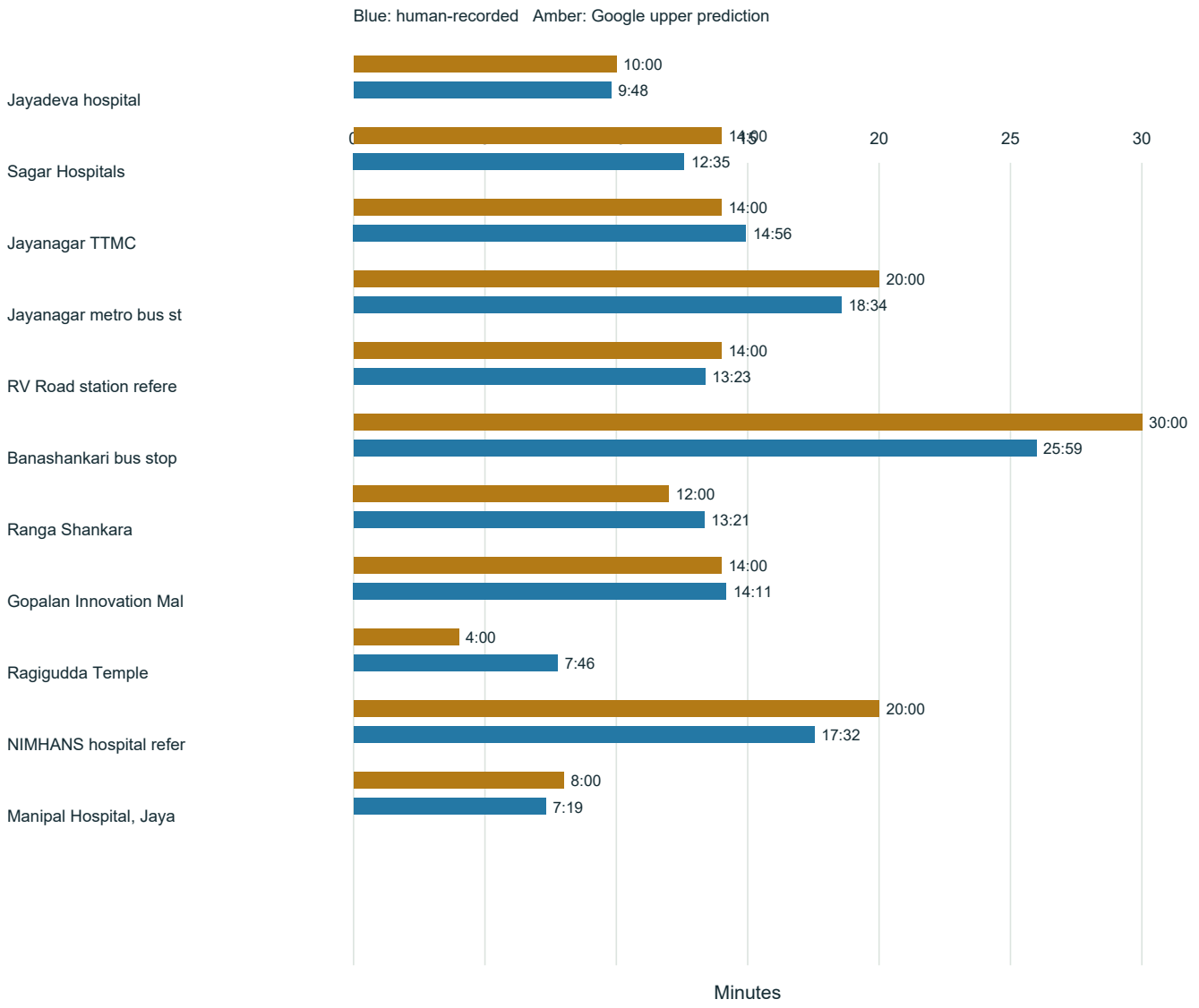
B is retained in the table because it is a recorded human trip, but its selected Google destination does not reliably match the intended bus-stop side and road level. It is excluded from the charts and aggregate count. For C-M excluding B: 7 of 11 human times are at or below the selected Google prediction; four are higher. The median difference is −39 seconds; the range runs from −4 min 01 sec to +3 min 46 sec. These descriptive differences do not identify the cause of a mismatch.

Why the comparison is limited

Trips can differ because of departure time, traffic, travel mode, path, endpoint, stop access, or rounding. Google incorporates traffic and route conditions into some forecasts, while the graph's OSRM profile times are static. The human observations are complete journeys, not measurements for each individual graph edge. A controlled comparison would record the same mode, route, start and end points, and repeated departures at matched times.

Phase 3 - Human and Google comparison graphs

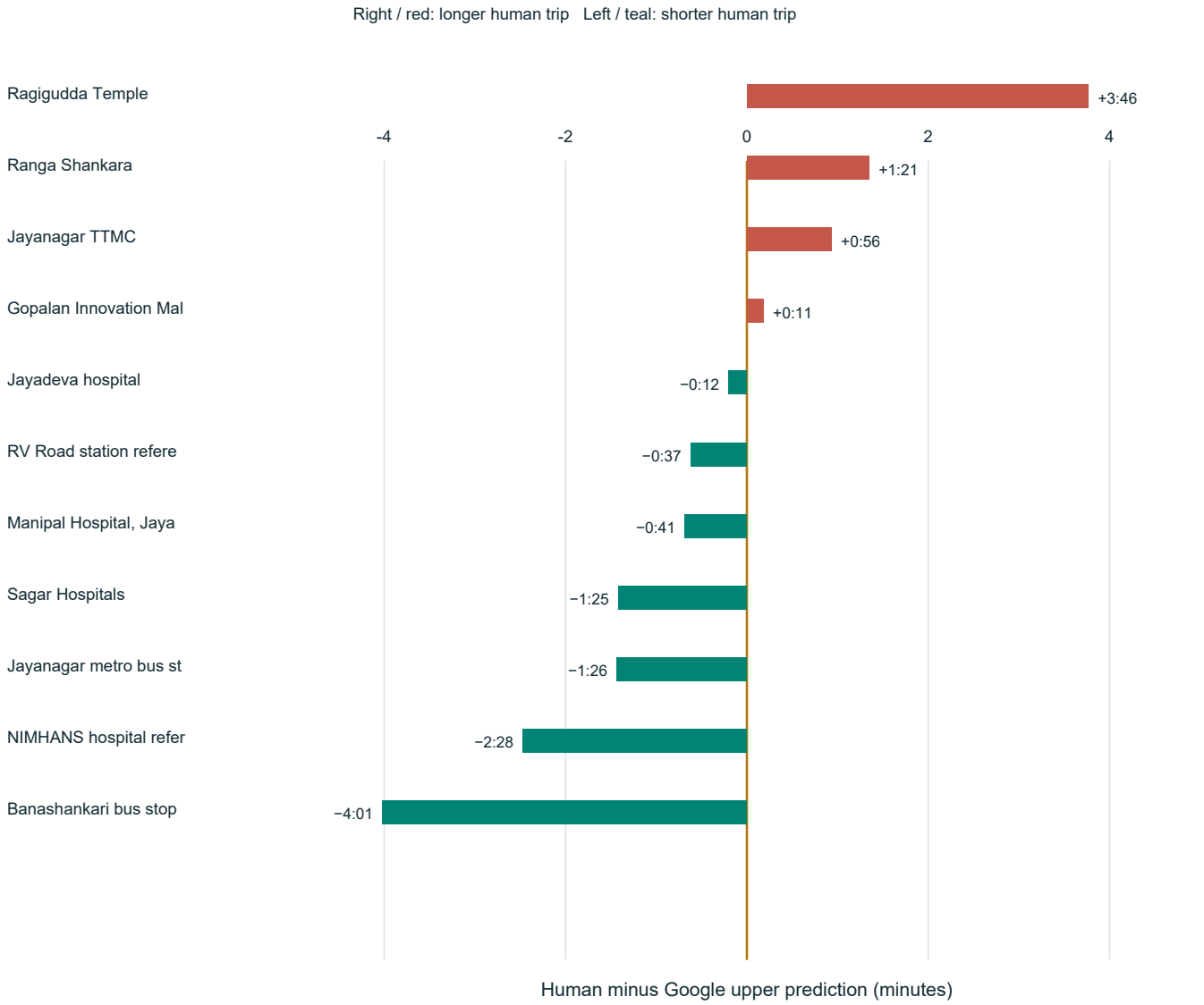
Paired durations. The comparison contains the 11 destinations whose Google endpoint is treated as comparable; B is left out.



Each pair shows the human-recorded trip and Google's selected upper prediction in minutes and seconds. The bars compare whole trips, not the graph's edge weights.

Phase 3 - Difference by destination

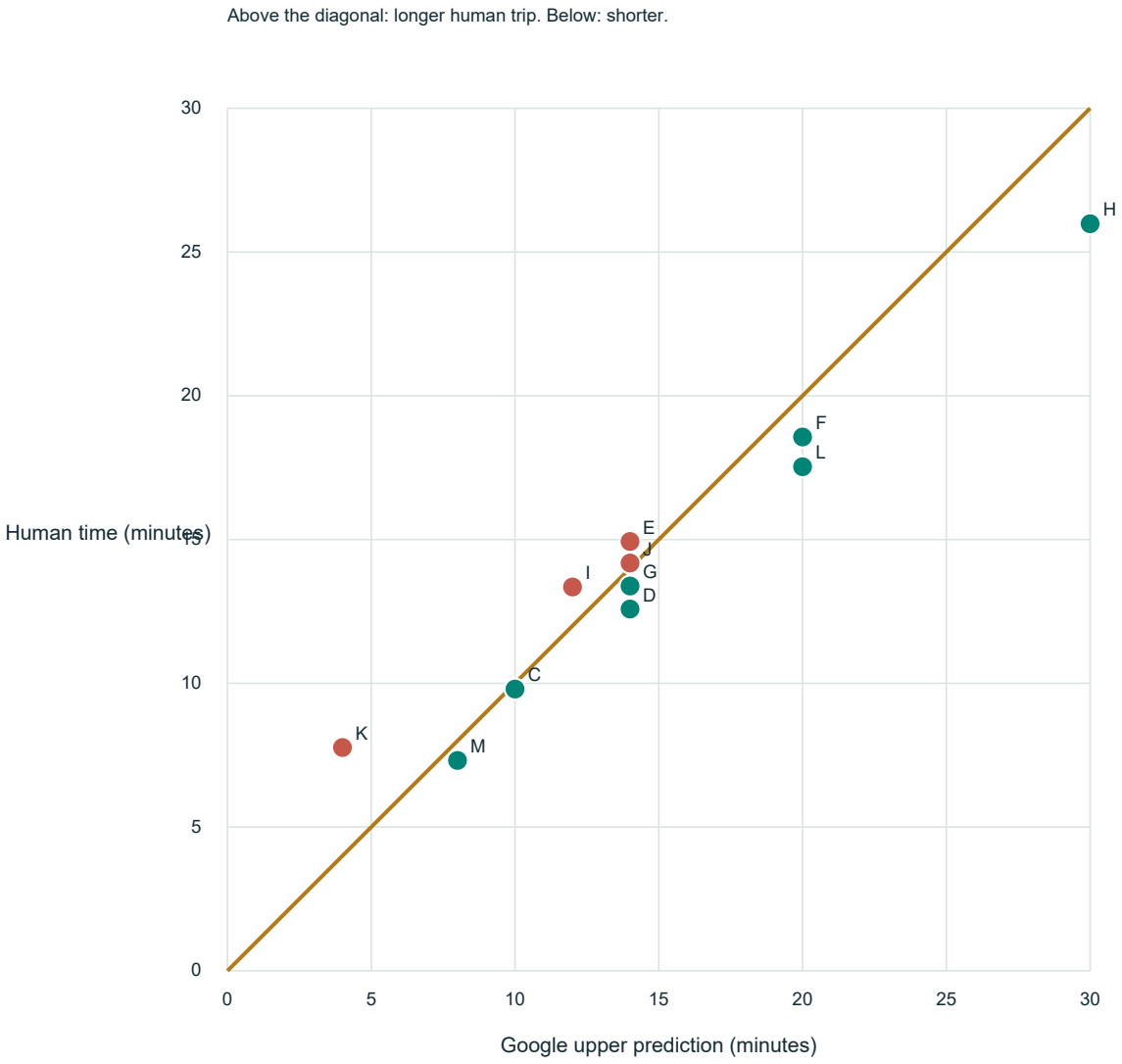
Difference is human-recorded duration minus Google’s upper prediction. Negative values mean the recorded trip was quicker; positive values mean it was longer.



The largest positive difference is Ragigudda Temple (K), +3 min 46 sec. The largest negative difference is Banashankari bus stop (H), −4 min 01 sec. The comparison does not show why those gaps occurred.

Phase 3 - Human time against Google prediction

Each point is one comparable destination. The diagonal represents equal durations. Points above it have longer human-recorded trips; points below it have shorter recorded trips.



The cluster around the equality line indicates similar rounded durations for several trips, while other destinations differ by several minutes. Google's selected upper estimate gives a cautious reference for this comparison; it is not a measured worst case.

Phase 3 - What traffic and emergency context add

Traffic in this study

Google’s selected busy-period upper estimate reflects a scheduled forecast and its traffic model. The static OSRM profile data used for the teaching graph do not update with current congestion. A future traffic experiment could capture Google’s low and high estimates, then repeat human drives and record time, route, mode, and weather at the same departure windows. The present data do not support attributing a difference to traffic alone.

Emergency-response interpretation

A CAG audit of Karnataka’s 108 ambulance records reported that 52.4% of analyzed road-accident responses arrived within 15 minutes and 15.03% took more than 30 minutes. Those administrative results describe a broad set of historical incidents; they are not Bengaluru-specific estimates for these destinations and do not validate this graph. [S4]

NHS England publishes category-specific response indicators, including separate mean and 90th-centile measures for the highest-priority category. This illustrates why a single route time cannot stand in for an emergency-service response measure. The definitions and service system differ from Bengaluru; they are methodological context only. [S5]

A complete response interval can include call handling, triage, dispatch, vehicle availability, mobilisation, travel, and handover. This graph models only travel between selected road references. It cannot establish improved ambulance response, clinical benefit, or lives saved.

Conclusion

The learning outcome is that Dijkstra produces a reproducible optimum only relative to a clearly defined graph and objective. The route can change when distance is replaced by time or when a corridor weight changes. The Google comparison shows that actual whole-trip durations can be either shorter or longer than a scheduled prediction, while still leaving route and condition differences unresolved. The strongest next study step is a controlled classroom route-choice exercise plus matched, repeated travel observations if the project intends to study traffic.

Phase 4 - 180-minute learning-session plan

The session uses prediction, manual tracing, code verification, controlled changes, and reflection. The human journey dataset is used as context; it is not presented as participants' route-choice or hand-calculation performance.

Minutes	Activity	Evidence to capture
0-15	Introduce source A, graph weights, objectives, and question	Individual route prediction
15-45	Inspect 13 locations, road snaps, and directed corridor weights	Annotated network and endpoint caveats
45-65	Collect independent route guesses before showing Dijkstra results	Chosen route and area familiarity; no timings claimed unless collected
65-95	Manually trace A to C and reconstruct predecessors	Trace table and route arithmetic
95-120	Run the algorithm and compare distance versus modeled time	Route outputs and code-time method
120-140	Remove BC or triple its duration; predict then rerun	Before/after route and cost
140-160	Read human-versus-Google charts and traffic caveats	Evidence-based comparison statement
160-180	Discuss limits and write an individual reflection	Exit response and one proposed improvement

If collecting a classroom sample

Use a consistent graph, weights, instructions, and destination tasks. Record participants' guessed path, objective, familiarity, task order, and route-choice time separately. If they manually calculate Dijkstra, record completion time separately from route-choice time. Hide the algorithm answer until guesses are recorded. Use several participants or counterbalanced task order to reduce learning effects. Do not treat the already supplied actual journey times as classroom experiment timings.

Suggested calculations

For a chosen route under a specified objective: route gap (%) = 100 × (chosen cost − optimal cost) ÷ optimal cost. An optimal choice has a 0% gap. Report the participant count, median gap, and proportion with an optimal route. Keep route distance, modeled travel duration, human journey duration, decision time, manual trace time, and code runtime in separate measures.

Printable learning worksheet

Participant code: _____ Task: C / D / L / M / J Objective: distance / modeled time

My predicted route from A: _____

Area familiarity: _____ Task order: _____ Decision time (if measured): _____

Initialize A=0 and all other tentative costs to infinity. At each step, settle the smallest tentative value, relax its outgoing edges, and note any improved labels and predecessors.

Step	Settled node	Cost	Improved labels / predecessors

Reconstructed route: _____

Chosen cost: _____ Optimal cost: _____ Route gap: _____

Reflection

1. Which edge changed your route prediction?
2. Did the shortest route differ from the fastest modeled route? Why?
3. Which missing real-world variable could change a trip?
4. Did the mapped endpoint match the intended stop or entrance?
5. What additional evidence would make a traffic comparison fairer?

Appendix C - Data definitions and source notes

S1 - JAIN University - campus location

Open source page

Source address used for A: 44/4 District Fund Road, Jayanagara 9th Block, Bengaluru.

S2 - OpenStreetMap contributors - map data and licence

Open source page

Mapped feature IDs, coordinates, and attribution are retained in the project source data. Licence: ODbL 1.0.

S3 - OSRM HTTP API - table and route data

Open source page

Table durations are seconds; route distances are metres along the selected profile routes. The public endpoint's exact profile build and map extract are not available in the saved response.

S4 - Comptroller and Auditor General of India - Karnataka 108 ambulance audit

Open source page

Primary audit source for the historical road-accident response-time figures. These are context only, not a route-model target.

S5 - NHS England - Ambulance Quality Indicators and AmbSYS specification

Open source page

Official indicator definitions and response-time statistics. The service categories and jurisdiction differ from Karnataka.

S6 - Manipal Hospitals - Jayanagar branch contact and address

Open source page

Official branch reference used to identify M.

S7 - Google Maps - route and busy-period predictions

Open source page

The report uses the saved upper end of the displayed driving forecast. Route and prediction snapshots may change; the report does not treat them as live data.

Terms used in the report

Teaching graph cost: sum of selected directed corridor weights. **Distance-optimal:** route with least graph distance. **Fastest modeled route:** route with least static profile duration. **Google prediction:** upper end of the saved busy-period Google driving forecast. **Difference:** human journey duration minus Google prediction. **Road snap:** distance from the mapped place coordinate to the routing engine's road reference.

Data completeness

The report contains the full location table, selected corridor weights, complete distance and duration matrices, graph outcomes, code timing measures, all simple route options for the four hospital targets, all hospital manual traces for both objectives, all human journey times, the Google upper predictions, difference calculations, controlled scenarios, and the learning materials. Human trip route, travel mode, departure time, repeat count, classroom route choices, route-choice time, and hand-calculation time were not supplied; these fields are left unclaimed.